

Salt Bridge Analysis — *Trametes versicolor* (Control)

Method

- Structure: ESMFold prediction of *Trametes versicolor* laccase (519 aa)
- Region of interest: Cupredoxin copper-binding loop, residues SER 82 - GLN 104 (feature activation 1.013 - 0.830)
- Tool: ChimeraX H-Bonds, "Salt bridges only" filter, distance tolerance 0.4 Å, angle tolerance 20 degrees, intramodel only, selection-limited search
- Search covered the full protein (not restricted to the loop residues only), so results below are the complete set found; those near/within the loop region are the ones relevant to the hypothesis

Validated Results (ChimeraX H-Bonds tool, official count)

Salt bridge	Contact atoms (count)	Distance range (Å)	Charge type
ARG 63 — ASP 116	3	2.679 - 3.132	Strong, unconditional
ARG 263 — ASP 444	3	2.661 - 3.180	Strong, unconditional
ARG 443 — ASP 244	3	2.857 - 3.099	Strong, unconditional
HIS 86 — ASP 444	1	2.845	Conditional (pH-dependent)
HIS 420 — ASP 97	1	2.786	Conditional (pH-dependent)

Total: 5 unique salt bridges (3 strong/unconditional, 2 conditional via Histidine)

Manual cross-check (raw distance method, for reference only)

A manual distance-only search (no angle validation) was performed first and found 17 candidate pairs under 4 Å, plus 6 near-misses beyond 4 Å. The manual method agreed closely with the validated tool on the ARG263-ASP444 pair (nearly identical distances), but included several pairs that did not survive proper angle validation. The ChimeraX H-Bonds tool result above is the accurate, citable count — the manual list is retained only as a process cross-check, not as the reported result.

Note on residue location relative to the loop

None of the 5 confirmed salt bridges have both residues falling directly inside the SER82-GLN104 loop range itself — the closest is ARG 63, which sits near but outside that exact range. This is worth stating plainly rather than smoothing over: the copper-binding loop, as narrowly defined by the feature-activation heatmap, may not itself be the region carrying the relevant salt bridges. The bridges found are elsewhere on the protein. This raises an open question about whether "near the copper-binding loop" needs to be defined more broadly (e.g. within the larger domain) before a meaningful candidate-vs-control comparison can be made.

Next step

Repeat the identical process on *Bacillus safensis* (candidate), loop range LEU 104 - PHE 125, using the same ChimeraX H-Bonds settings, for a direct, apples-to-apples comparison.